

JSS1 Biology Diagnostic Bank (40 Questions)

Scope: Primary 5/6 Foundation & Full JSS1 Cambridge/Nigerian Biology

Focus: Life Science Literacy & JSS2 Readiness

Section 1: Characteristics & Classification (Questions 1 - 15)

No.	Question & Options	Ans	Source & Diagnostic Purpose
1	Which mnemonic is commonly used to remember the characteristics of living things? A) MR NIGER D B) PEMDAS C) ROYGBIV D) SMART	A	[P5/6] Foundational recall of life characteristics.
2	The process by which living things produce offspring is: A) Growth B) Reproduction C) Excretion D) Nutrition	B	[P5] Identification of life processes.
3	A cat running after a mouse is an example of which life characteristic? A) Respiration B) Movement/Irritability C) Nutrition D) Growth	B	[JSS1] Application of MRS GREN/MR NIGER D to real scenarios.
4	Living things are classified into groups. The largest group is called a: A) Species B) Genus C) Kingdom D) Family	C	[JSS1/Cambridge] Introduction to Taxonomy.
5	Which of these belongs to the Kingdom Plantae? A) Mushroom B) Moss C) Amoeba D) Yeast	B	[JSS1] Distinguishing plants from fungi. Common misconception check.
6	Vertebrates are animals with: A) Hair B) Backbones C) Wings D) Gills	B	[P5/6] Basic classification of animals.
7	An example of an invertebrate is: A) Snake B) Frog C) Snail D) Bird	C	[P6] Applying classification rules.
8	Which group of animals has moist skin and lays eggs in water? A) Reptiles B) Amphibians C) Mammals D) Fish	B	[P6/JSS1] Characteristics of vertebrate classes.
9	All living things are made up of basic units called: A) Tissues B) Atoms C) Cells D) Organs	C	[JSS1/Cambridge] Transitioning to cellular biology.
10	Plants that produce seeds in flowers are called: A) Angiosperms B) Gymnosperms C) Ferns D) Algae	A	[JSS1] Botanical classification.
11	Fungi are different from plants because they cannot: A) Grow B) Reproduce C) Photosynthesize D) Breathe	C	[JSS1] Kingdom differences (Nutrition source).
12	Micro-organisms that cause diseases are known as: A) Parasites B) Pathogens C) Predators D) Preys	B	[JSS1] Health science terminology.
13	A key used to identify organisms based on characteristics is a: A) Map B) Dichotomous Key C) Ruler D) Compass	B	[Cambridge JSS1] Practical biological tools.
14	Which of these is a unicellular organism? A) Human B) Tree C) Amoeba D) Fish	C	[JSS1] Complexity of life forms.
15	The scientific name of an organism consists of its: A) Kingdom and Phylum B) Genus and Species C) Order and Family D) Class and Genus	B	[JSS1/Cambridge] Binomial nomenclature basics.

Section 2: Cells & Human Systems (Questions 16 - 30)

No.	Question & Options	Ans	Source & Diagnostic Purpose
16	Which part of the cell is the "brain" or control center? A) Cytoplasm B) Cell Wall C) Nucleus D) Vacuole	C	[JSS1] Basic cell anatomy.
17	Plant cells have a _____ which animal cells lack. A) Cell Membrane B) Nucleus C) Cell Wall D) Cytoplasm	C	[P6/JSS1] Distinguishing plant and animal cells.
18	The jelly-like substance where chemical reactions happen in a cell is: A) Vacuole B) Cytoplasm C) Nucleus D) Ribosome	B	[JSS1] Cell component functions.
19	A group of similar cells working together is a: A) Organ B) Tissue C) System D) Organism	B	[P6/JSS1] Levels of organization.
20	The heart, arteries, and veins make up the: A) Nervous System B) Circulatory System C) Digestive System D) Respiratory System	B	[P5] Human body systems identification.
21	Digestion begins in the: A) Stomach B) Small Intestine C) Mouth D) Esophagus	C	[P5/6] Digestive process flow.
22	The organ responsible for removing carbon dioxide from the body is: A) Liver B) Kidneys C) Lungs D) Skin	C	[P6/JSS1] Excretory/Respiratory functions.
23	Which part of the skeletal system protects the brain? A) Ribs B) Spine C) Skull D) Pelvis	C	[P5] Skeletal system functions.
24	Muscles are attached to bones by tough cords called: A) Ligaments B) Tendons C) Nerves D) Joints	B	[P6/JSS1] Support and movement.
25	The sensory organ for balance is the: A) Eye B) Ear C) Nose D) Skin	B	[P6] Sensory functions.
26	Puberty is the stage where: A) A baby starts walking B) The body becomes capable of reproduction C) A person stops growing D) Teeth fall out	B	[JSS1] Human development/Reproductive health.
27	Balanced diet must contain how many main food groups? A) 3 B) 5 C) 7 D) 6	D	[P5/6] Fundamental nutrition.
28	Iodine is used to test for the presence of: A) Protein B) Starch C) Fat D) Sugar	B	[JSS1/Cambridge] Laboratory testing in Biology.
29	The chemical that makes plants green is: A) Blood B) Chlorophyll C) Sap D) Water	B	[P5/6] Plant biology foundations.
30	Exchange of gases in plants occurs through tiny pores called: A) Roots B) Stomata C) Veins D) Bark	B	[JSS1] Plant physiology.

Section 3: Ecology & Health (Questions 31 - 40)

No.	Question & Options	Ans	Source & Diagnostic Purpose
31	The place where an organism lives is its: A) Niche B) Habitat C) Community D) Population	B	[P5/6] Basic ecological terms.
32	In a food chain, green plants are always: A) Consumers B) Decomposers C) Producers D) Predators	C	[P6] Energy flow in ecosystems.
33	An animal that eats only plants is a: A) Carnivore B) Omnivore C) Herbivore D) Parasite	C	[P5] Feeding habits.
34	Several overlapping food chains form a: A) Food Web B) Pyramid C) Circle D) Line	A	[JSS1] Ecosystem complexity.
35	Pollution of the air by smoke and harmful gases causes: A) Global warming B) Soil erosion C) Flooding D) Earthquakes	A	[JSS1/Cambridge] Environmental science.
36	Abstinence is a way to avoid: A) Fever B) Unwanted pregnancy and STIs C) Hunger D) Tiredness	B	[JSS1] Health and social education.
37	Which of these is a non-communicable disease? A) Malaria B) Diabetes C) Flu D) Tuberculosis	B	[JSS1] Health classification.
38	Water can be made safe for drinking by: A) Adding salt B) Boiling and Filtering C) Cooling D) Stirring	B	[P5/6] Personal health and sanitation.
39	Which of these increases the rate of soil erosion? A) Planting trees B) Deforestation C) Mulching D) Terracing	B	[P6/JSS1] Impact of human activity on the environment.
40	Final Diagnostic Note: Students scoring high in Sections 1 & 2 show strong aptitude for Clinical Sciences/Medicine pathways.	-	[Final Note] Pathway readiness indicator.